Strict global PF_4 for the Riemann kernel and a sharp Fourier separator

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Abstract

Let Φ be the classical Riemann, or de Bruijn–Newman, kernel. We prove that every translation minor of Φ of order at most four is strictly positive at strictly ordered nodes, and a direct origin-confluent calculation gives a negative order-five minor. Thus the exact global Polya-frequency order is four. The computer-assisted input is confined to directed one-variable inequalities: global positivity of the log-curvature quantities q, F_2 and of the fully confluent invariant $\mathcal{C}_4$, together with a five-derivative origin certificate using exact rational arithmetic. The remainder is exact. A quotient-Wronskian argument reduces arbitrary fourth-order minors to $\partial_\xi \Psi < 0$. In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w W(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2} \, \mathrm{d}t.$$

Here W is the difference of two explicitly normalized triangular CDFs. Its strict positivity follows from a one-crossing density ratio. All load-bearing algebraic reductions are displayed in the text and appendices; the repository scripts provide replay rather than substitute derivations. We then construct an explicit positive even Schwartz kernel which is itself strictly PF_4 but whose entire Fourier transform has nonreal zeros. Thus continuous PF_3 and PF_4 membership alone do not force Fourier real-rootedness. The results are finite-order and have no implication for the Riemann Hypothesis.

Contents

1 Introduction

A function $f : \mathbb{R} \to \mathbb{R}$ is PF_r when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^{k}$$

is nonnegative for $1 \le k \le r$ and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Polya-frequency order of the Riemann kernel; see [1] for modern background and Schoenberg's characterization.

Theorem 1.1 (Strict global PF_4)**.** *For every $1 \le k \le 4$ and strictly increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^{k} > 0.$$

A parity factorization of the fully confluent derivative matrix at the origin gives a negative order-five limit. Divided differences then transfer that sign to equally spaced translation minors at sufficiently small positive spacing; see Theorem 10.2. Hence Theorem 1.1 has the following sharp consequence.

Corollary 1.2. *The exact global Polya-frequency order of Φ is four.*

The classification is sharp as a condition on Fourier zeros, even within the positive even Schwartz class.

Theorem 1.3 (Continuous PF_4 Fourier separator)**.** *There is an explicit positive even Schwartz function f_* which is strict PF_4, while its entire Fourier transform has nonreal zeros. Consequently continuous PF_3 and continuous PF_4 membership are each insufficient to force Fourier real-rootedness.*

The proof has two certified one-variable inputs and one exact transport mechanism. First, global inequalities $q > 0$ and $F_2 > 0$ imply strict PF_3 and positivity of a three-point functional Λ. Second, the fully confluent order-four invariant $\mathcal{C}_4$ is globally positive. Exact quotient identities reduce strict PF_4 to the monotonicity of a scalar three-point function Ψ. An increasing curvature coordinate then turns the derivative of Ψ into a positive crossing-kernel integral of $\mathcal{C}_4$.

The distinction between a human derivation and a computer certificate is maintained throughout. Directed interval arithmetic proves the compact and tail inequalities stated in Section 3; exact algebra is written in the paper and independently replayed by the repository verifiers. The short order-five certificate uses only exact rational arithmetic and elementary series bounds. The classification concerns finite-order total positivity only. It neither proves nor disproves the Riemann Hypothesis. Theorem 1.3, proved in Section 11, makes the finite-order boundary explicit rather than leaving it as a logical caveat.

2 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 x}, \qquad \Theta(t) = \vartheta(e^{2t}), \qquad H(t) = e^{t/2}\Theta(t).$$

The theta transformation $\vartheta(x) = x^{-1/2}\vartheta(1/x)$ gives

$$\Theta(-t) = e^t\Theta(t), \qquad H(-t) = H(t).$$

The series and all its derivatives converge locally uniformly, so H is real analytic and even. Define

$$\Phi(t) = \frac{1}{2}\left(H''(t) - \frac{1}{4}H(t)\right).$$

Termwise differentiation gives, for every real t,

$$\Phi(t) = \sum_{n\geq 1}\left(4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2}\right) e^{-\pi n^2 e^{2t}}. \tag{2.1}$$

Thus the positive-side formula often written with $|t|$ is the restriction of a real-analytic even function. In particular, smoothness at the origin is not inferred from numerical coverage.

Write

$$\ell = \log\Phi, \qquad s = \ell', \qquad q = -s' = -\ell''.$$

For $a < b$, define

$$A(a,b) = s(a) - s(b), \qquad M(a,b) = \frac{q(b) - q(a)}{A(a,b)}. \tag{2.2}$$

When $q > 0$, s is strictly decreasing and $A(a,b) > 0$.

3 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \qquad F_2 = q^3 - F_1.$$

For the logarithmic cumulants $c_j = \ell^{(j)}$, set

$$\begin{aligned} &m_0 = 1, \quad m_1 = 0, \quad m_2 = c_2, \quad m_3 = c_3, \\ &m_4 = c_4 + 3c_2^2, \quad m_5 = c_5 + 10c_2c_3, \\ &m_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3, \end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \tag{3.1}$$

Lemma 3.1 (Certified input)**.** *For the kernel* Φ*,*

$$q(t) > 0, \qquad F_2(t) > 0, \qquad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The compact interval $[0, 1]$ is covered with directed Arb balls at 192-bit precision. The retained lower bounds are

$$q \geq 18.7268, \qquad F_2 \geq 3889.2, \qquad \mathcal{C}_4 \geq 2.817 \times 10^7.$$

The first certificate uses 7731 adaptive second-order Taylor cells and the second 8050. Evenness covers $[-1, 1]$.

For $t \geq 1$, put $x = \pi e^{2t}$. The cumulants have directed enclosures

$$c_j = -2^j x + \varepsilon_j, \qquad |\varepsilon_j| \leq \frac{E_j}{x}, \qquad 2 \leq j \leq 6,$$

where

$$(E_2, E_3, E_4, E_5, E_6) = (19.8, 176, 2082, 30770, 545900).$$

They imply $q, F_2 > 0$ on the tail. Substitution in the determinant (3.1) gives

$$\frac{\mathcal{C}_4(t)}{x^6} > 44392 \qquad (x \geq \pi e^2).$$

The derivation of the E_j, the exact lower polynomial, and its monotonic normalized margin are recorded in Appendix B. These are the only directed numerical inputs to the main positive proof.

4 Order three and the positive slope functional

For $\xi < m < r$, define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \tag{4.1}$$

We give the full estimate used to prove its positivity.

Set $f_0 = q'/q$. Because $s' = -q$,

$$M(a, b) = \frac{\int_a^b q'(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t} = \frac{\int_a^b f_0(t) q(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t};$$

thus $M(a, b)$ is a q-weighted mean of f_0. Choose a point $u \in [\xi, m]$ where f_0 is minimal on the left interval and a point $v \in [m, r]$ where it is maximal on the right. Then

$$M(m, r) - M(\xi, m) \leq f_0(v) - f_0(u) \leq \int_\xi^r \max(f_0'(t), 0)\,\mathrm{d}t.$$

Since

$$f_0' = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and $A(\xi, r) = \int_\xi^r q(t)\,\mathrm{d}t$, equation (4.1) yields

$$\Lambda(\xi; m, r) \geq \int_\xi^r \left(q - \frac{\max(F_1, 0)}{q^2} \right) \mathrm{d}t \tag{4.2}$$

$$= \int_\xi^r \frac{q^3 - \max(F_1, 0)}{q^2}\,\mathrm{d}t = \int_\xi^r \frac{\min(q^3, F_2)}{q^2}\,\mathrm{d}t > 0. \tag{4.3}$$

The last inequality follows from Lemma 3.1 and $q > 0$.

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (4.3), together with strict log-concavity, proves strict global PF_3. We rederive the needed Wronskian form in the next section because its signs also control order four.

5 Quotient-Wronskian reduction

Fix $y_1 < y_2 < y_3 < y_4$, put $u_j(t) = \Phi(t - y_j)$, and write $p_j = t - y_j$, so $p_4 < p_3 < p_2 < p_1$. Define

$$v_j = (u_j/u_1)', \qquad w_j = (v_j/v_2)'.$$

Successive column division and differentiation give

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \tag{5.1}$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{5.2}$$

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. From (2.2),

$$\mathcal{T} A(a, b) = q(b) - q(a), \qquad \mathcal{T} \log A(a, b) = M(a, b). \tag{5.3}$$

This sign is load-bearing. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2}\frac{A(p_j, p_1)}{A(p_2, p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j, p_2) + M(p_j, p_1) - M(p_2, p_1). \tag{5.4}$$

The exact weighted-mean split

$$A(p_j, p_1) M(p_j, p_1) = A(p_j, p_2) M(p_j, p_2) + A(p_2, p_1) M(p_2, p_1)$$

turns (5.4) into

$$g_j = \frac{A(p_j, p_2)}{A(p_j, p_1)} \Lambda(p_j; p_2, p_1) > 0. \tag{5.5}$$

Thus $w_j = (v_j/v_2) g_j > 0$, and (5.1) has the strict order-three sign.

Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r).$$

Because $w_j = (v_j/v_2)g_j$,

$$\mathcal{T}\log(w_4/w_3) = (g_4 + \mathcal{T}\log g_4) - (g_3 + \mathcal{T}\log g_3). \tag{5.6}$$

Equation (5.5) and (5.3) give

$$\mathcal{T}\log g_j = M(p_j, p_2) - M(p_j, p_1) + \mathcal{T}\log\Lambda_j,$$

where $\Lambda_j = \Lambda(p_j; p_2, p_1)$. A second use of the same mean split gives

$$g_j + M(p_j, p_2) - M(p_j, p_1) = \Lambda_j - A(p_2, p_1).$$

The common last term cancels in (5.6), yielding the exact identity

$$\mathcal{T}\log(w_4/w_3) = \Psi(p_4; p_2, p_1) - \Psi(p_3; p_2, p_1). \tag{5.7}$$

Proposition 5.1 (Three-point equivalence)**.** *Under the strict lower-order positivity already proved, global* PF_4 *is equivalent to*

$$\partial_\xi \Psi(\xi; m, r) \le 0 \qquad (\xi < m < r).$$

Strict inequality implies strict global PF_4.

Proof. If $\partial_\xi \Psi \le 0$, then $p_4 < p_3$ makes the right side of (5.7) nonnegative. All prefactors in (5.2) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For increasing row nodes $t_1 < \cdots < t_k$, divide columns by $u_1(t_i)$ and take consecutive row differences from the bottom upward. The fundamental theorem of calculus and determinant multilinearity give

$$\det[U_j(t_i)]_{i,j=1}^{k} = \int_{t_1}^{t_2} \cdots \int_{t_{k-1}}^{t_k} \det[U'_{j+1}(\tau_i)]_{i,j=1}^{k-1}\, \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_{k-1},$$

where $U_1 = 1$. Repeating the quotient reduction converts the Wronskian sign to every collocation minor; positivity of the integrand on a product of positive-length intervals gives the strict conclusion.

Conversely, assume global PF_4. Let all four row nodes coalesce, divide by their positive Vandermonde, and pass to the limit. The resulting translate Wronskian is nonnegative. Equations (5.2) and (5.7) imply $\Psi(p_4; p_2, p_1) \ge \Psi(p_3; p_2, p_1)$ whenever $p_4 < p_3$. Thus $\Psi(\cdot; m, r)$ is nonincreasing, which is equivalent to the stated differential inequality because it is smooth. $\square$

6 Curvature coordinate and sign bridge

Since $q > 0$,

$$y(t) = -s(t), \qquad Q(y(t)) = q(t)$$

defines a strictly increasing global coordinate, with $\mathrm{d}y/\mathrm{d}t = q(t) = Q(y) > 0$. Primes in this and the next three sections mean y-derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \qquad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For $\xi < m < r$, write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because $A(a,b) = y(b) - y(a)$ and $M(a,b) = Q[y(a), y(b)]$,

$$\Lambda = (w - p) + Q[p,z] - Q[z,w]. \tag{6.1}$$

Two integrations of $\kappa = 2 - Q''$ give

$$\Lambda = \frac{1}{L}\int_p^z (y-p)\kappa(y)\,\mathrm{d}y + \frac{1}{R}\int_z^w (w-y)\kappa(y)\,\mathrm{d}y, \tag{6.2}$$

$$\delta := -\partial_p\Lambda = \frac{1}{L^2}\int_p^z (z-y)\kappa(y)\,\mathrm{d}y > 0. \tag{6.3}$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p)\partial_p + Q(z)\partial_z + Q(w)\partial_w,$$

and $\partial_\xi = Q(p)\partial_p$. Thus $\Lambda_\xi = -Q(p)\delta$.

Define

$$\mathcal{N} = \delta\Lambda^2 + Q'(p)\delta\Lambda + \Lambda\mathcal{T}\delta - \delta\mathcal{T}\Lambda. \tag{6.4}$$

Differentiating $\Psi = \Lambda + \mathcal{T}\log\Lambda$, commuting simultaneous translation with ∂_ξ, and collecting over Λ^2 gives

$$\partial_\xi\Psi = -\frac{Q(p)}{\Lambda^2}\mathcal{N}. \tag{6.5}$$

Equivalently,

$$\frac{\mathcal{N}}{\delta\Lambda} = K := \Lambda + Q'(p) + \mathcal{T}\log\delta - \mathcal{T}\log\Lambda. \tag{6.6}$$

It remains to prove $K > 0$.

7 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log\kappa)'\}'. \tag{7.1}$$

Lemma 7.1 (Confluent-to-curvature identity)**.** *The determinant* (3.1) *satisfies*

$$\mathcal{C}_4 = Q^6\kappa^2 D. \tag{7.2}$$

Consequently $D > 0$ on $\mathbb{R}$.

Proof. The change of variable gives $\mathrm{d}/\mathrm{d}t = Q\,\mathrm{d}/\mathrm{d}y$ and $c_2 = -Q$. Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (3.1) factors out Q^6. The remaining factor is

$$\kappa^2\left(3(\kappa-1) - \{Q(\log\kappa)'\}'\right),$$

as shown by the common expanded numerator in Appendix A. This proves (7.2) without a numerical assumption. Lemma 3.1, $Q > 0$, and $\kappa > 0$ then imply $D > 0$. □

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by Φ^4. The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.

8 Exact transport identity

Normalize the triangular weights in (6.2) and (6.3) as probability measures:

$$\mathrm{d}\mu(y) = \frac{(z-y)\kappa(y)}{L^2\delta}\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y, \tag{8.1}$$

$$\mathrm{d}\nu(y) = \frac{(y-p)\kappa(y)}{L\Lambda}\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y + \frac{(w-y)\kappa(y)}{R\Lambda}\mathbf{1}_{[z,w]}(y)\,\mathrm{d}y. \tag{8.2}$$

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y)\frac{\kappa'(y)}{\kappa(y)}.$$

Then $A_0' = D$.

Lemma 8.1 (Transport cancellation)**.** *With K as in* (6.6)*,*

$$K = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \tag{8.3}$$

Proof. Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = H', \qquad H = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \tag{8.4}$$

$$H = J', \qquad J = y^3 - 3yQ + QQ'. \tag{8.5}$$

Both identities follow by one differentiation.

Let $I_L = J(z) - J(p)$ and $I_R = J(w) - J(z)$. Integrating the linear weights in (8.1)–(8.2) by parts once gives

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{LH(p) - I_L}{L^2\delta}. \tag{8.6}$$

On the other hand, differentiation of the triangular integrals along $\mathcal{T}$ gives

$$\mathcal{T}\Lambda = U(z,w) - U(p,z), \quad \mathcal{T}\delta = \frac{U(p,z) - (Q(z) - Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d)-Q(c))^2}{(d-c)^2}. \tag{8.7}$$

Substitution in (6.6) yields

$$K = \Lambda + Q'(p) - Q[p,z] + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)\kappa(p)}{L\delta}. \tag{8.8}$$

The endpoint identity equating (8.6) and (8.8), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of Q, Q', Q''; no bound or polynomial model is used. This proves (8.3). □

9 The positive crossing kernel

Let F_μ, F_ν be the CDFs of (8.1)–(8.2), and put

$$W(y) = F_\mu(y) - F_\nu(y).$$

Lemma 9.1 (Strict stochastic ordering)**.** *One has $W(y) > 0$ for $p < y < w$, while $W(p) = W(w) = 0$.*

Proof. On $[z, w]$, $F_\mu = 1$, so $W = 1 - F_\nu > 0$ before w. On (p, z), the ratio of the two positive densities is

$$R_0(y) = \frac{\mathrm{d}\mu/\,\mathrm{d}y}{\mathrm{d}\nu/\,\mathrm{d}y} = \frac{\Lambda(z-y)}{L\delta(y-p)}. \tag{9.1}$$

Writing $C = \Lambda/(L\delta) > 0$, direct differentiation gives

$$R_0'(y) = -\frac{C(z-p)}{(y-p)^2} < 0. \tag{9.2}$$

Thus R_0 decreases strictly from $+\infty$ to 0. Hence $W' = \mu' - \nu'$ changes sign exactly once, from positive to negative. Since $W(p) = 0$ and

$$W(z) = 1 - F_\nu(z) = \nu((z, w]) > 0,$$

the function cannot return to zero on $(p, z]$. The right interval was already handled. $\square$

Integration by parts for CDFs, using $A_0' = D$ and the common total mass, turns Lemma 8.1 into

$$K = \int_p^w W(y)D(y)\,\mathrm{d}y. \tag{9.3}$$

Combining (6.6), (7.2), and (9.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w W(y)\frac{\mathcal{C}_4(y)}{Q(y)^6\kappa(y)^2}\,\mathrm{d}y > 0. \tag{9.4}$$

Every factor in the interior is strictly positive.

10 Completion and exact order

Equation (6.5) and (9.4) imply

$$\partial_\xi\Psi(\xi; m, r) < 0 \qquad (\xi < m < r).$$

Proposition 5.1 therefore proves Theorem 1.1.

We now prove directly that the classification stops at order four. Put

$$H_k = \det[\Phi^{(i+j)}(0)]_{i,j=0}^{k-1}, \qquad s_j = \Phi^{(j)}(0).$$

Since Φ is even, simultaneous parity permutations of the rows and columns give

$$H_4 = AB, \qquad H_5 = CB,$$

where

$$A = s_0 s_4 - s_2^2, \qquad B = s_2 s_6 - s_4^2,$$

and

$$C = \det\begin{pmatrix} s_0 & s_2 & s_4 \\ s_2 & s_4 & s_6 \\ s_4 & s_6 & s_8 \end{pmatrix}.$$

Lemma 10.1. *For the Riemann kernel,*

$$445 < A < 446, \qquad 189223 < B < 189228, \qquad -674000000 < C < -614000000.$$

In particular, $H_4 > 0$ *and* $H_5 < 0$.

Proof. For $x = \pi n^2 e^{2u}$, the n-th summand of $\Phi(u)$ is $e^{u/2-x}P_0(x)$, where $P_0(x) = 2x(2x-3)$. If its j-th derivative is $e^{u/2-x}P_j(x)$, then

$$P_{j+1} = 2xP_j' + (1/2 - 2x)P_j.$$

Thus

$$s_j = \sum_{n\geq 1} e^{-\pi n^2} P_j(\pi n^2).$$

The first three modes are enclosed by rational alternating-series bounds. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ supplies rational bounds for π, and argument division followed by the degree-29 and degree-30 alternating Taylor sums supplies rational bounds for each exponential.

For the omitted modes, let S_j be the sum of the absolute coefficients of P_j. Since $\deg P_j = j+2$, $3 < \pi < 22/7$, and $e^3 > \sum_{r=0}^{8} 3^r/r! > 20$,

$$\sum_{n\geq 4} |P_j(\pi n^2)| e^{-\pi n^2} \leq \frac{S_j (22/7)^{j+2} 4^{2j+4} 20^{-16}}{1-(5/4)^{20}20^{-9}} \qquad (j = 0, 2, 4, 6, 8).$$

Exact rational interval arithmetic then gives the stated bounds. The full endpoints and the standard-library verifier are included in the reproducibility archive. □

Theorem 10.2. *The Riemann kernel is not* PF_5.

Proof. For a C^{2k-2} function f, repeated row and column divided differences give, for increasing fixed tuples a, b,

$$\lim_{h\downarrow 0} \frac{\det[f(z + h(a_i - b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)} V(a) V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1} (r!)^2} \det[f^{(i+j)}(z)]_{i,j=0}^{k-1}.$$

Take $k = 5$, $z = 0$, and $a_i = b_i = i$. The sign factor is one, both Vandermonde factors are positive, and Lemma 10.1 makes the limit strictly negative. Hence $\det[\Phi((i-j)h)]_{i,j=0}^{4} < 0$ for every sufficiently small positive h. These are translation minors at strictly increasing nodes. □

Theorem 1.1 and Theorem 10.2 prove Corollary 1.2.

The proof uses no multidimensional interval cover of the nonconfluent configuration space. Earlier positive-tail Peano densities, collision cones, Hermite boxes, and escape charts remain independent certified results, but they are not premises here. Their role was superseded when the transport numerator was recognized as the difference of expectations in (8.3).

Remark 10.3 (Riemann Hypothesis boundary). The theorem determines the kernel's exact finite Polya-frequency order and makes no claim about the zeros of the Riemann Ξ-function. In particular, the failure of PF_5 neither proves nor disproves the Riemann Hypothesis. The next section proves, by an explicit strict-PF_4 separator, that no Fourier real-rootedness conclusion follows from membership through order four alone.

11 A sharp continuous PF_4 separator

We now prove Theorem 1.3. Put

$$c = \frac{1}{64}, \qquad B(w) = \sum_{k=-3}^{3} b_k w^k, \qquad (b_{-3}, \dots, b_3) = (2, 10, 23, 30, 23, 10, 2),$$

and define

$$f_*(x) = e^{-cx^2} B(e^{2cx}) = \sum_{k=-3}^{3} b_k e^{ck^2} e^{-c(x-k)^2}. \tag{11.1}$$

The Gaussian-mixture expression proves positivity and Schwartz decay; palindromy of (b_k) proves evenness.

11.1 Strict PF_3

For fixed x, give $K \in \{-3, \dots, 3\}$ probability proportional to $b_k e^{2ckx}$, and write κ_j for its cumulants. Differentiation of the finite log-partition function gives

$$q = 2c - (2c)^2\kappa_2, \qquad q' = -(2c)^3\kappa_3, \qquad q'' = -(2c)^4\kappa_4. \tag{11.2}$$

Since the support interval has length six, Popoviciu's inequality yields $0 \le \kappa_2 \le 9$, and hence

$$q \ge 2c(1 - 18c) = 2c\frac{23}{32} > 0. \tag{11.3}$$

Moreover,

$$\begin{aligned} F_2 &= q^3 - \{qq'' - (q')^2\} \\ &= q^3 + q(2c)^4\kappa_4 + (2c)^6\kappa_3^2. \end{aligned} \tag{11.4}$$

The identity $\kappa_4 = \mathbb{E}(K - \mathbb{E}K)^4 - 3\kappa_2^2$ gives $\kappa_4 \ge -243$. Dropping the final nonnegative term in (11.4) gives

$$F_2 \ge q\{q^2 - 3888c^4\}.$$

At $c = 1/64$, the two coefficients after division by c^2 are

$$q^2/c^2 \ge \frac{529}{256}, \qquad 3888c^2 = \frac{243}{256}.$$

Thus $F_2 \ge qc^2(143/128) > 0$ globally. The weighted-mean criterion of Section 4 proves strict PF_3.

11.2 Exact order-four density

Let $\epsilon = 2c = 1/32$, $t = e^{\epsilon x}$, and

$$P(t) = t^3 B(t) = 2 + 10t + 23t^2 + 30t^3 + 23t^4 + 10t^5 + 2t^6.$$

For $E = t\frac{\mathrm{d}}{\mathrm{d}t}$, define

$$N_1 = tP', \qquad N_{j+1} = t(N_j'P - jN_jP'). \tag{11.5}$$

The quotient rule gives $E^j \log P = N_j/P^j$. Therefore, for $\ell_j = (\log f_*)^{(j)}$,

$$\ell_2 = \frac{-\epsilon P^2 + \epsilon^2 N_2}{P^2}, \qquad \ell_j = \frac{\epsilon^j N_j}{P^j} \quad (3 \le j \le 6). \tag{11.6}$$

Insert these jets into the central-moment determinant (3.1). Every monomial has derivative weight twelve, so the common denominator is P^{12}:

$$\mathcal{C}_4[f_*](x) = \frac{H(t)}{P(t)^{12}}. \tag{11.7}$$

The exact expansion in Appendix E proves that H is palindromic of degree 72 and that all 73 coefficients are strictly positive. Hence $\mathcal{C}_4[f_*] > 0$ on $\mathbb{R}$.

Strict PF_3 also gives

$$\kappa = 2 - Q'' = \frac{q^3 + F_2}{q^3} > 0.$$

The generic transport identity (9.4) now has strictly positive density and crossing weight for f_*. Thus $\partial_\xi \Psi < 0$, and Proposition 5.1 proves that f_* is strict PF_4.

11.3 Nonreal Fourier zeros

With $\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-izx}\,\mathrm{d}x$, Gaussian translation gives

$$\widehat{f}_*(z) = \sqrt{\frac{\pi}{c}} e^{-z^2/(4c)} \sum_{k=-3}^{3} b_k e^{ck^2} e^{-ikz}. \tag{11.8}$$

Put $w = e^{-iz}$ and $u = w + w^{-1}$. The Laurent factor in (11.8) vanishes exactly when

$$R_c(u) = 2e^{9c}u^3 + 10e^{4c}u^2 + (23e^c - 6e^{9c})u + 30 - 20e^{4c} \tag{11.9}$$

vanishes. At $c = 1/64$, directed 192-bit arithmetic gives

$$\mathrm{disc}(R_c) \in -6.0446137251764984704554480068690598 + [-3.14, 3.14]10^{-34},$$

which is strictly negative. Thus the real cubic has a nonreal conjugate pair. For $|w| = 1$, however, $w + w^{-1}$ is real. The corresponding w-roots therefore lie off the unit circle and lift through $w = e^{-iz}$ to zeros with $\operatorname{Im} z \neq 0$. The Gaussian factor is zero-free. This completes the proof of Theorem 1.3.

12 Reproducibility and evidence boundary

The paper has six replay boundaries.

Object	Repository verifier	Role
Global q, F_2, strict PF_3	`scripts/verify_pf3_reduction.py`	directed and exact
Global $\mathcal{C}_4 > 0$	`scripts/verify_c4_confluent.py`	directed and exact
Quotient reduction	`scripts/verify_pf4_wronskian_reduction.py`	exact generic algebra
Transport completion	`scripts/verify_pf4_transport_kernel.py`	exact algebra and checks
Origin order-five obstruction	`scripts/verify_pf5_origin.py`	exact rational
Continuous strict-PF_4 separator	`scripts/verify_continuous_pf4_separator.py`	exact and directed

The compact inequalities are directed Arb calculations at 192-bit precision; the tail uses explicit analytic envelopes and rational coefficient bounds. The origin order-five certificate is independent: it uses only standard-library integer arithmetic with outward rounding on a rational grid, Machin's formula, alternating Taylor sums, three retained theta modes, and one geometric tail bound. The quotient, curvature, primitive, endpoint, density-ratio, and confluent-limit identities are mathematical identities for generic smooth functions. The paper displays these identities; the scripts independently replay them.

A focused independent audit should proceed in this order:

1. verify the analytic theta normalization in (2.1);
2. replay the directed $q, F_2, \mathcal{C}_4$ compact and tail certificates;
3. check the sign $\mathcal{T} \log A = M$ and the two weighted-mean splits in Section 5;
4. expand the determinant and curvature identity in Appendix A;
5. check the endpoint identity in Appendix C;
6. differentiate the density ratio in (9.2);
7. derive the parity factorization and divided-difference limit in Section 10, then replay the rational origin certificate;
8. reconstruct the separator's central determinant; then verify the coefficient signs and Fourier discriminant in Appendix E.

No numerical scan is used as evidence for global nonconfluent positivity of either kernel.

A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det \begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 3 and ordinary three-by-three expansion gives

$$\begin{aligned} \mathcal{C}_4 = {} & 12c_2^6 + 24c_2^4 c_4 - 24c_2^3 c_3^2 + 2c_2^3 c_6 - 12c_2^2 c_3 c_5 + 7c_2^2 c_4^2 \\ & + 12c_2 c_3^2 c_4 + c_2 c_4 c_6 - c_2 c_5^2 - 9c_3^4 - c_3^2 c_6 + 2c_3 c_4 c_5 - c_4^3. \end{aligned}$$

In the curvature coordinate, repeated application of $Q\,\mathrm{d}/\,\mathrm{d}y$ to $c_2 = -Q$ gives

$$\begin{aligned}
c_2 &= -Q,\\
c_3 &= -QQ',\\
c_4 &= -Q\{(Q')^2 + QQ''\},\\
c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\},\\
c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}.
\end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned}
\mathcal{P} = {} & QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q'''\\
& + 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12.
\end{aligned}$$

Now $\kappa = 2 - Q''$, and direct differentiation gives

$$3(\kappa - 1) - \{Q(\log\kappa)'\}' = -\frac{\mathcal{P}}{(Q''-2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore $\mathcal{C}_4 = Q^6\kappa^2 D$, proving (7.2) by an exposed common numerator.

B Tail envelopes and normalized margin

For $x = \pi e^{2t}$, write

$$\Phi(t) = 4\pi x e^{5t/2-x}(1+\rho(x)), \qquad \rho(x) = -\frac{3}{2x} + \sigma(x), \qquad w = \log(1+\rho).$$

The $n \geq 2$ theta terms satisfy, with their first six derivatives, an exponentially small bound absorbed into the following rational paddings for $x \geq \pi e^2$:

$$|w^{(k)}(x)| \leq \frac{C_k}{x^{k+1}},$$

k	1	2	3	4	5	6
C_k	1.61	3.33	10.4	42.8	222	1376

For $D_x = 2x\,\mathrm{d}/\,\mathrm{d}x$, induction gives the Stirling identity

$$D_x^j\psi = 2^j\sum_{k=1}^{j} S(j,k)x^k\psi^{(k)},$$

where $S(j,k)$ are Stirling numbers of the second kind. Since only w contributes to the error in derivatives of order at least two,

$$|\varepsilon_j| \leq \frac{E_j}{x}, \qquad E_j = 2^j\sum_{k=1}^{j} S(j,k)C_k.$$

Rounding each sum upward gives

j	2	3	4	5	6
E_j	19.8	176	2082	30770	545900.

For example,

$$E_4 \leq 16(C_1 + 7C_2 + 6C_3 + C_4) = 2081.92 < 2082.$$

The logarithmic derivative formulas for w are finite Faa di Bruno sums; the $n = 1$ contribution uses $\rho' \sim 3/(2x^2), \rho'' \sim -3/x^3, \ldots$, while the common theta-tail padding controls σ and all its derivatives. The verifier evaluates these inequalities with directed balls.

Substituting $c_j = -2^j x + \delta_j E_j/x$, $|\delta_j| \leq 1$, into the thirteen-term polynomial and collecting absolute coefficients gives

$$\begin{aligned} P(x) = 49152x^6 - \frac{7299072}{5}x^4 - \frac{88018944}{5}x^3 - \frac{756929024}{5}x^2 - \frac{9307871232}{25}x \\ - \frac{90509073984}{25} - \frac{1807239327472}{125x} - \frac{3528006311808}{125x^2} - \frac{62513985514124}{625x^3} \\ - \frac{220885105972212}{3125x^4} - \frac{8406623961648}{625x^5} - \frac{11297761792812}{15625x^6}. \end{aligned}$$

Every subtracted term in $P(x)/x^6$ has the form $b_d x^{d-6}$ with $b_d > 0$ and $d < 6$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{P(x)}{x^6} = \sum_{d<6}(6-d)b_d x^{d-7} > 0.$$

The normalized margin is increasing, so its minimum on $x \geq \pi e^2$ occurs at the left endpoint. Evaluation at the rational lower bound $x > 1157/50$ gives

$$\frac{\mathcal{C}_4}{x^6} \geq \frac{P(x)}{x^6} > 44392.80919.$$

C Endpoint audit of the transport cancellation

This appendix records the algebra suppressed by the phrase "simplifying the endpoint expression." Put

$$z = p + L, \qquad w = z + R, \qquad a = \frac{Q(z) - Q(p)}{L}, \qquad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of $2 - Q''$ gives

$$\delta = \frac{L + Q'(p) - a}{L}, \qquad \Lambda = L + R + a - b. \tag{C.1}$$

Define

$$J_y = y^3 - 3yQ(y) + Q(y)Q'(y),$$
$$H_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d - c} + \frac{(Q(d) - Q(c))^2}{(d-c)^2}.$$

With $I_L = J_z - J_p$ and $I_R = J_w - J_z$, the identity needed in Lemma 8.1 is exactly

$$\frac{-(J_z - J_p)/L + (J_w - J_z)/R}{\Lambda} + \frac{LH_p - (J_z - J_p)}{L^2\delta}$$
$$= \Lambda + Q'(p) - a + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)(2 - Q''(p))}{L\delta}.$$

To check it, substitute (C.1), replace z by $p+L$ and w by $p+L+R$, and multiply by the common denominator $L^2 R\delta\Lambda$. The terms containing $Q'(w)$, then $Q'(z)$, then $Q''(p)$ cancel pairwise; the remaining endpoint values and position terms cancel after inserting $a = (Q(z) - Q(p))/L$ and $b = (Q(w) - Q(z))/R$. No differential equation for Q is used, so the identity holds for arbitrary smooth Q.

D Proof provenance

The maintained source map is `paper/PAPER_MAP.md`. The principal repository proof records are:

Claim	Proof record
Global q, F_2, order three	`sources/pf3-global-certificate.md`
Global confluent $\mathcal{C}_4$	`sources/c4-confluent-certificate.md`
Quotient and three-point equivalence	`sources/pf4-wronskian-reduction-certificate.md`
Transport identity and completion	`sources/pf4-transport-kernel-certificate.md`
Origin order-five obstruction	`sources/pf5-origin-certificate.md`
Continuous strict-PF$_4$ Fourier separator	`sources/continuous-pf4-separator-certificate.md`

The MIND identities are respectively $R142$–$R144$, $R147$–$R152$, $R154$–$R156$, $R153$, $R164$, $R14$, $R37$, $R50$, $R72$, and $R165$–$R170$, with replay boundaries $CERT2, CERT3, CERT5, CERT9, CERT11, CERT10$. These identifiers are evidence anchors, not bibliographic citations.

E Exact separator audit

This appendix records the algebraic boundary behind (11.7). Starting from the cumulants $\ell_2, \ldots, \ell_6$, the verifier reconstructs the central moments

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = \ell_2, \quad m_3 = \ell_3,$$

$$m_4 = \ell_4 + 3\ell_2^2, \qquad m_5 = \ell_5 + 10\ell_3\ell_2,$$

$$m_6 = \ell_6 + 15\ell_4\ell_2 + 10\ell_3^2 + 15\ell_2^3,$$

and then forms $\det[m_{i+j}]_{i,j=0}^3$ directly. This independently regenerates the thirteen-term $\mathcal{C}_4$ polynomial; it does not import the expansion used for the Riemann kernel.

The recurrence (11.5) is then applied over $\mathbb{Q}[t]$. Substitution of (11.6) shows that each of the thirteen determinant monomials has common denominator P^{12}. After collecting the numerator H, exact rational arithmetic verifies

$$\deg H = 72, \qquad [t^j]H = [t^{72-j}]H > 0 \quad (0 \le j \le 72),$$

with

$$\min_{0\le j\le 72}[t^j]H = \frac{3}{65536}.$$

Coefficient positivity is a proof on the full domain $t > 0$, not a sampled test. Exact spot reconstructions at $t = 1/3, 1, 4$ independently compare the central determinant with H/P^{12}.

For the Fourier step, the identities

$$w + w^{-1} = u, \quad w^2 + w^{-2} = u^2 - 2, \quad w^3 + w^{-3} = u^3 - 3u$$

give (11.9) directly. The only directed numerical step is the 192-bit Arb enclosure of its discriminant. All remaining checks are symbolic or exact rational arithmetic.

References

[1] A. Belton, D. Guillot, A. Khare, and M. Putinar, *Preservers of totally positive kernels and Polya frequency functions*, arXiv:2110.08206.